# Congress is Non-binary: A Mixed Membership Exploration of the 118th Congress

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## Abstract

In contemporary American politics, national and local politics are considered in popular consciousness to be on a 1-dimensional left-right, Democrat-Republican divide. In our current reality and historical tradition of a two-party system in the United States, this left-right dimension often does separate the parties in the modern House of Representatives. However, given the different blocs that are in the modern legislature, this paper seeks to answer whether the voting behavior and floor speeches of representatives can help distinguish various factions within (or across) parties.

We construct a vote-agreement network of 428 representatives from 360 final-passage roll calls and model it with a degree-corrected mixed-membership model, estimated with Mixed-SCORE. This model was extended with covariates derived from embeddings of the respective member's floor speeches in the Congressional Record. The resulting eigenvalue spectrum indicates four communities: progressive Democrats, moderate Democrats, mainstream Republicans, and Freedom Caucus Republicans.

The memberships of these groups are strongly mixed: only 7.5% of members place more than 0.9 of their weight on a single community, and 83 of 210 Democrats and 68 of 218 Republicans place more than a quarter of their weight on communities dominated by the other party. Weighting each member's speech by membership and scoring words with a log-odds ratio with an informative Dirichlet prior shows that communities also differ in what they discuss. These contrasts persist within parties after party-wide vocabulary is canceled out.

## 1. Introduction

The two-party system has long been a hallmark of American politics, with the Democrats and Republicans serving as the most well-known parties, and many politicians firmly associating themselves with either or. But is the distinction between beliefs actually that clear? Can our politics really be reduced to a simple Democrat/Republican distinction? In this paper, we seek to explore whether or not politicians' voting behavior & congressional speeches adds another dimension to consider, hence, is Congress non-binary?

To start, we begin by inspecting whether or not there might be other communities at hand in the House, and how mixed representatives' memberships are. We note that members often don't belong to just one faction, but rather, sit between many of them. To do this, we conduct mixed-membership community detection by modeling House members with a degree-corrected mixed-membership (DCMM) model, in which each member has a membership vector $\pi_i$ over K communities. We estimate $\pi_i$ with Mixed-SCORE (Jin et al., 2024), a spectral method, extended with node covariates following Ji et al. (2022).

We then inspect the words that each community uses to analyze if there are any trends and notable distinctions between topics that they tend to talk about more on the House floor. To do this, we start with some basic natural language processing and cleaning, before reweighting so that each representative has equal weight in the process and no single representative skews their community's word frequencies. We then weight community word counts with soft membership, where each member's words are split across communities in proportion to their membership weights $\pi_i$.

## 2. Methodology

### 2.1. Data

#### 2.1.1. Speeches

Floor speeches were sourced from the daily edition of the Congressional Record, the official and publicly available account of the proceedings and debate of the United States Congress, published by the Government Publishing Office and accessed through GovInfo. They were extracted with downloading and parsing code adapted from the boundaries_congress package (van der Does & Kolchinsky, 2021), which (Zhao, 2025) also used. This paper analyzes the 118th Congress.

Zhao's pipeline filtered speeches heavily: procedural speakers were removed, speeches had to be at least 250 characters long and have a document title containing "Act" or "Resolution," and the remaining speeches were matched to bills by title. In the 117th Congress this retained 10,485 of about 96,747 speeches (about 11%). A further coverage filter on speakers and topics left 134 legislators, and the thesis notes that this selection may overrepresent members of leadership and committee chairs.

Because the present analysis models the whole House, a lighter selection was used:

- House speeches with a member identifier (bioguide ID)
- At least 250 characters
- Dated before November 1 of the election year, which avoids using information from after the election

In the 118th Congress, this selection reduced 97,817 speeches (House and Senate) to 28,914. Among the 428 members of the network, 427 have at least five retained speeches and 368 (86%) have at least twenty, with a median of 47 speeches per member. Participation in floor speech remains uneven, so some selection toward more active speakers is expected in the member-level speech features. About one third of all rows have no member identifier, mostly presiding officers and similar roles. The Record for January 3, 2023 (the opening day of the 118th Congress) is unavailable from GovInfo.

Each retained speech was embedded with BAAI/bge-small-en-v1.5 (384 dimensions). Speeches longer than 300 words were split into chunks (at most six per speech), chunk embeddings were averaged with weights proportional to word count, and speech embeddings were averaged to the member level, again weighted by word count (Xiao et al., 2023). Principal components of the member-level vectors were fit on the 115th and 116th Congresses and applied to the 118th Congress. In the training Congresses, the first five components explain 19.1%, 6.5%, 5.0%, 4.4%, and 4.0% of the variance. The leading component correlates with the share of Extensions of Remarks (magnitude 0.67 to 0.75 across the 115th, 116th, and 118th Congresses) and only weakly with party (0.12 in the 118th); example speeches indicate that it separates tribute and recognition speeches from floor debate. The third component relates more strongly to party (0.45 in the 118th). The covariates include Extensions of Remarks, while the word analysis excludes them.

#### 2.1.2. Voting Record

To generate a graph which represents voting information, we first pulled the voting records from the Voteview database (Lewis et al., 2025). This allowed us to attain a record of representatives and how they voted in a congress. We focus on the 118th Congress and use only final passage roll calls on bills and joint resolutions (360 votes).

We then built an undirected weighted graph where two representatives are connected with a weight corresponding to how similarly they voted. For instance if two congress people cast the same vote on $x$ number of votes and cast different votes on $y$ votes they would be connected with a weight of $S_{ij} = \frac{x}{x+y}$. Here only votes on which both representatives cast a Yea or Nay vote are counted with absences ignored. Every pair of representatives is connected with no self connection. We call this matrix, $S$, the agreement matrix.

We require representatives to have voted on at least half of the bills (180 of 360) to be included, to avoid noisy and poorly behaved nodes, since a representative with few votes has agreement weights based on only a few shared bills. This leaves 428 of the 445 representatives who served in the 118th Congress. The adjacency weight matrix is included as figure 1 in the appendix.

### 2.2. Mixed Membership Model

We first decided on how many communities to try and detect. To do this we looked at the absolute values of the eigenvalues of the agreement matrix. Looking at the scree plot, figure 2, we see that after the fourth eigenvalue the other eigenvalues level off and do not appear to carry significant signal. We therefore decided to run a mixed membership community detection with $K = 4$ communities.

We use our weighted matrix $S$ in place of $A$ in the Mixed-SCORE algorithm (Jin et al., 2024) for the DCMM model, where $\Omega_{ij} = \theta_i\theta_j\pi_i^\top P\pi_j$ and $E[A] = \Omega - \mathrm{diag}(\Omega)$. Since $S_{ij}$ is the frequency with which congress person $i$ and $j$ vote the same way, it is an estimate of $P(i \text{ votes the same as } j)$, and we treat it as an approximation of $E[A_{ij}] = \Omega_{ij}$. This means that $S$ is a nearly noiseless approximation of $E[A]$. Let $\hat{\lambda}_k$ and $\hat{\xi}_k$ denote the $k$-th eigenvalue and eigenvector of $S$, ordered by absolute value. Taking the 4 leading eigenvectors of $S$, we performed the SCORE algorithm (Jin, 2015), which with $K = 4$ gives

$$\hat{R} = \left[\frac{\hat{\xi}_2}{\hat{\xi}_1}, \quad \frac{\hat{\xi}_3}{\hat{\xi}_1}, \quad \frac{\hat{\xi}_4}{\hat{\xi}_1}\right]$$

We then added in embeddings of speeches each congressional member had made as covariate information. After standardizing the embedding's 5 PCs, we took the top

$K = 4$ left singular vectors, call these $\hat{\eta}_1, \hat{\eta}_2, \hat{\eta}_3, \hat{\eta}_4$, and define

$$\hat{H} = \left[\hat{\eta}_1, \quad \hat{\eta}_2, \quad \hat{\eta}_3, \quad \hat{\eta}_4\right].$$

We then rescaled $\hat{H}$ so its total variability across members (sum of squared robust standard deviations of columns) is the same as that of $\hat{R}$. We used the coefficient $c_{\text{scale}} = 13.9$. Then we used $\tau = \tau_0 \cdot c_{\text{scale}}$, where $\tau_0$ is the relative weight tuning parameter. We used $\tau_0 = 1$ (equal weight), giving $\tau = 13.9$. This gives

$$Z = \left[\hat{R},\ \tau\hat{H}\right].$$

Once we have $Z$, we use the sketched vertex search, running $k$-means with $10 \cdot K = 40$ clusters and then applying the successive projection algorithm to the cluster centers to find the $K$ vertices $\hat{v}_1, \ldots, \hat{v}_K$. We then expressed each member as a convex combination of these vertices using barycentric weighting, solving for each member $i$

$$\hat{w}_i = \arg \min_{w \geq 0,\ \sum_k w_k = 1} \left\| z_i - \sum_{k=1}^{K} w_k \hat{v}_k \right\|^2,$$

where $z_i$ is the $i$-th row of $Z$. Finally we rescaled these weights to account for the degree parameters, $\hat{\pi}_i^*(k) = \hat{w}_i(k)/\hat{b}_1(k)$ with $\hat{b}_1(k) = \left[\hat{\lambda}_1 + \hat{v}_k^{R\top} \text{diag}(\hat{\lambda}_2, \ldots, \hat{\lambda}_K)\hat{v}_k^R\right]^{-1/2}$, where $\hat{v}_k^R$ is the first $K-1$ coordinates of $\hat{v}_k$ (the part corresponding to $\hat{R}$), and normalized to obtain $\hat{\pi}_i = \frac{\hat{\pi}_i^*}{\|\hat{\pi}_i^*\|_1}$ so each member's weights sum to one.

### 2.3. Speech Analysis

Our goal for the word analysis was to see whether the communities found by the mixed membership model differ in what they talk about on the House floor. To do this, we wanted a list of words that each community uses more than the others, which we could then visually represent as word clouds. To start, we thought about how we would go about turning speeches into word counts, how to give each community respective word counts when there are members that belong to several communities, and how to decide which differences in word use are meaningful.

We began by building one document per member by combining all of their floor speeches in the 118th Congress. Starting from the speeches described in Section 2.1.1, we also dropped the Extensions of Remarks, which are written statements inserted into the Record rather than spoken on the floor. This left 20,513 speeches from 426 of the 428 members in our network. For data cleaning purposes, we then lowercase adjusted the text and kept words of at least three letters. Besides common English stopwords, we removed words tied to floor procedure rather than content (e.g. "yield," "Madam Speaker," "reclaiming my time," "amendment"), party names, and state names. To keep a representative's pet words or hometown out of the results, we kept only words used by at least 10 members, which gave a vocabulary of 12,506 words.

Let $c_{iw}$ be the number of times member $i$ says word $w$, and $L_i = \sum_w c_{iw}$ their total word count. We also wanted to account for the fact that different members of the House differ a lot in how often they speak, and we didn't want a few outstandingly verbose individuals to skew a community's word counts. To make sure that each member was given equal weight, we kept those 373 members with $L_i \geq 1000$ and rescaled each member's counts to the median length $\tilde{L}$,

$$\tilde{c}_{iw} = c_{iw} \times \frac{\tilde{L}}{L_i}$$

To account for the mixed membership of representatives, when accounting for their word count, we weighted them accordingly to each community in proportion with their membership. For example, for two groups $A$ and $B$ with member weights $a_i$ and $b_i$, we have that the group word counts are

$$y_{A,w} = \sum_i a_i \tilde{c}_{iw}, y_{B,w} = \sum_i b_i \tilde{c}_{iw}$$

with totals $n_A = \sum_w y_{A,w}$ and $n_B = \sum_w y_{B,w}$. Then, to compare community $m$ with the rest of the House, we solved where: $a_i = \hat{\pi}_i(m)$ and $b_i = 1 - \hat{\pi}_i(m)$. To find differences within a certain party, we took the two communities $m$ and $n$ that hold the most weight among that party's members, and set $a_i = \hat{\pi}_i(m)$ and $b_i = \hat{\pi}_i(n)$ for members of that party, and $a_i = b_i = 0$ for all other members. Because both groups then consist only of members of the same party, words used heavily by the whole party appear on both sides and cancel, so the remaining differences reflect divisions within the party (and therefore the specific communities we found) rather than the difference between Democrats and Republicans. Accounting for the fact that analyzing simple raw frequencies would lead to greater issues (since common words like "people" are going to just be on everyone's list), and that a ratio of frequencies would also be unsound (as rare words would then just dominate the list), we therefore employ the weighted log-odds ratio with an informative Dirichlet prior (Monroe et al., 2008). With prior counts $\alpha_w = 0.01 \sum_i \tilde{c}_{iw}$ and $\alpha_0 = \sum_w \alpha_w$,

$$\hat{\delta}_w = \log \frac{y_{A,w} + \alpha_w}{n_A + \alpha_0 - y_{A,w} - \alpha_w} - \log \frac{y_{B,w} + \alpha_w}{n_B + \alpha_0 - y_{B,w} - \alpha_w},$$

$$z_w = \hat{\delta}_w \Big/ \sqrt{\frac{1}{y_{A,w} + \alpha_w} + \frac{1}{y_{B,w} + \alpha_w}}.$$

The prior shrinks words with few uses toward zero, and dividing by the SE means a frequent word needs a real difference in use to score highly. A positive $z_w$ means group $A$ uses the word more, and a negative $z_w$ means group $B$ does. As a final check, we dropped any word for which a single member supplies more than 40% of either group's weighted count.

We treated words with $|z_w| > 1.96$ as distinctive. Each word cloud shows a group's 40 words with the largest $z_w$, with word size proportional to $z_w$, so large words are the ones most reliably associated with that group rather than the ones it says most often.

#### 2.3.1. Validation and Interpretation Metrics

To interpret the communities, we compare them with DW-NOMINATE (Poole & Rosenthal, 1985; 1997), a standard measure of legislators' ideology. It is estimated from every roll-call vote in a Congress, and its first dimension runs from about -1 (most liberal) to +1 (most conservative). It is obtained from Voteview (Lewis et al., 2025).

We refer to $z_w$ as the Fightin' Words z-score (Monroe et al., 2008). It measures how much more one group uses word $w$ than another, in standard-error units. $z_w = 0$ means no difference, and a positive value means group A uses the word more. This metric prioritizes reliability of differences rather instead of favoring words that are frequent or rare. While we have set a threshold of $|z_w| > 1.96$, we use this more as a ranking mechanism than a significance test.

## 3. Results

### 3.1. The Communities

| | Size | Dem. | NOM. | Nearest members |
|---|---|---|---|---|
| V1 Progressive D | 103 | 81% | −0.25 | Bush, Pressley, Tlaib |
| V2 Moderate D | 106 | 80% | −0.18 | Peltola, Levin, Peters |
| V3 Mainstream R | 174 | 21% | +0.30 | Lucas, Latta, Wagner |
| V4 Freedom Caucus R | 45 | 14% | +0.52 | Roy, Biggs, Crane |

*Table 1.* The four communities ($K = 4$, $\tau_0 = 1$). Size is $\sum_i \hat{\pi}_{ik}$ (total 428). Dem. share and NOMINATE (first dimension) are $\hat{\pi}$-weighted averages. Nearest members have the largest $\hat{\pi}_{ik}$.

Table 1 summarizes the four different communities. Communities V1 and V2 are majority Democratic (81% and 80% of their weight comes from Democrats) and have effective sizes of 103 and 106. Communities V3 and V4 are majority Republican (21% and 14% Democratic) with effective sizes of 174 and 45. The $\hat{\pi}$-weighted mean NOMINATE scores of the four communities are $-0.25$, $-0.18$, $+0.30$, and $+0.52$, respectively. The members with the largest weight on each community are Bush, Pressley, Ramirez, Tlaib, and Omar (V1); Peltola, Levin, Vasquez, Peters, and Panetta (V2); Lucas, Latta, LaTurner, Graves, and Wagner (V3); and Roy, Biggs, Rosendale, Crane, and Good (V4).

### 3.2. Distribution of Memberships

Figure 4 plots each member's membership vector $\hat{\pi}_i$ in the tetrahedron, where the corners represent the four communities. We note that few members are close to a single corner: 7.5% of members have a largest weight $\max_k \hat{\pi}_i(k)$ above 0.9, and 78.3% have a largest weight above 0.5. The median largest weight is 0.64.

Assigning each member to the community with their largest weight, 108 Democrats fall in V1, 92 in V2, 10 in V3, and none in V4. Among Republicans, none fall in V1, 4 in V2, 184 in V3, and 30 in V4. On average, Democrats place 0.40 of their weight on V1, 0.40 on V2, 0.17 on V3, and 0.03 on V4; Republicans place 0.09 on V1, 0.10 on V2, 0.63 on V3, and 0.18 on V4. In total, 83 of 210 Democrats place more than 25% of their weight on the two Republican-majority communities (V3 and V4), and 68 of 218 Republicans place more than 25% on the two Democratic-majority communities (V1 and V2). Among Republicans, the difference $\hat{\pi}_i(4) - \hat{\pi}_i(3)$ has a correlation of 0.69 with NOMINATE.

### 3.3. Distinctive Words by Community

The word analysis covers the 373 members with at least 1,000 words of floor speech. Comparing each community with the rest of the House, the number of words with $z_w > 1.96$ (used more by the community) is 762 for V1, 407 for V2, 750 for V3, and 564 for V4. Table 2 lists the ten words with the largest $z_w$ for each community, and Figure 5 shows the top 40 as word clouds placed at the corners of the tetrahedron.

The highest-scoring words are "black" ($z = 26.8$), "gun," and "violence" for V1; "water" ($z = 13.0$), "oil," and "clean" for V2; "administration" ($z = 14.4$), "communist," and "Biden" for V3; and "spending" ($z = 15.6$), "trillion," and "Biden" for V4. "Biden" and "border" are among the top ten words of both V3 and V4.

### 3.4. Within-Party Differences

Among Democrats, the two communities with the most weight are V1 and V2; among Republicans, they are V3 and V4. Table 4 lists the words with the largest positive and negative $z_w$ for each comparison.

Among Democrats, 188 words have $z_w > 1.96$ (used more by V1) and 189 have $z_w < -1.96$ (used more by V2). The words most associated with V1 are "black" ($z = 11.7$), "people," "rights," "justice," and "immigrants"; those most

| Rank | V1 | V2 | V3 | V4 |
|---|---|---|---|---|
| 1 | black (26.8) | water (13.0) | administration (14.4) | spending (15.6) |
| 2 | gun (15.3) | oil (10.8) | communist (12.7) | trillion (15.2) |
| 3 | violence (15.1) | clean (10.5) | biden (12.5) | biden (14.6) |
| 4 | rights (12.2) | pollution (9.1) | border (11.3) | border (13.4) |
| 5 | people (12.2) | community (8.7) | chinese (11.0) | aliens (11.9) |
| 6 | extreme (10.9) | communities (8.1) | ccp (10.4) | illegal (11.4) |
| 7 | women (10.9) | las (8.1) | energy (10.0) | money (11.0) |
| 8 | health (10.4) | lands (8.1) | china (10.0) | government (11.0) |
| 9 | maga (10.3) | climate (7.9) | suspend (9.3) | administration (10.9) |
| 10 | social (10.1) | wildfires (7.2) | enforcement (9.1) | radical (9.7) |

*Table 2.* Top ten words used more by each community than by the rest of the House ($K = 4$, single words), with Fightin' Words $z$-scores in parentheses.

associated with V2 are "las" ($z = -5.6$), "coast," "water," "Hamas," and "military."

Among Republicans, 236 words have $z_w > 1.96$ (used more by V3) and 373 have $z_w < -1.96$ (used more by V4). The words most associated with V3 are "critical" ($z = 5.7$), "financial," "proud," "energy," and "community"; those most associated with V4 are "trillion" ($z = -10.7$), "people," "money," "spending," and "government."

## 4. Discussion

### 4.1. What the Results Suggest

The results point towards the idea that Congress is indeed non-binary. Notably, although party is the dominant structure in the House, it is far from being the only one. The second eigenvalue of the agreement matrix ($|\hat{\lambda}_2| = 75.0$) is roughly ten times larger than the third (7.0), indicating that the division between parties is substantially stronger than other patterns in the data. However, that said, the third and fourth eigenvalues suggest that these smaller divisions within parties are not simply random noise, and the resulting communities align with recognizable political factions in the 118th Congress.

Each party naturally splits into a mainstream and a wing, but the two splits clearly differ between the two. The Democratic communities, V1 and V2, are nearly equal in size (103 and 106) and close in NOMINATE ($-0.25$ and $-0.18$). The members nearest V1 (Bush, Pressley, Tlaib, Omar) are associated with the progressive wing of the party, while those nearest V2 (Peltola, Levin, Peters, Panetta) are generally regarded as more moderate. On the Republican side, the mainstream community V3 is large (174), while V4 is small (45) and lies farther to the right ($+0.30$ versus $+0.52$). Its nearest members (Roy, Biggs, Rosendale, Crane, Good) are prominent members of the House Freedom Caucus. The Republican split thus resembles a large mainstream with a small, distinct wing, while the Democratic split is more even.

Furthermore, the membership vectors support the idea that, for the most part, members tend to sit between factions rather than inside them. Only 7.5% of members are close to a single community. Moreover, the typical member places about two thirds of their weight on one community and spreads the rest across others. There's also some that place substantial weight on communities dominated by the other party (83 of 210 Democrats and 68 of 218 Republicans place more than a quarter of their weight there). This further supports our point that a simply binary narrative undermines Congress and fails to recognize its nuances.

Going one step further, the word analysis indicates that these communities differ not only in how they vote, but also the issues that they tend to discuss the most. Relative to the rest of the House, for example, we note that V1's discussions center on civil rights and social policy matters ("black," "gun violence," "rights," "health"), while V2's centers on the environment, energy, and local concerns ("water," "oil," "clean," "wildfires," "firefighters"). V3 emphasizes China and national security ("communist," "CCP," "China"), along with energy, while V4 emphasizes federal spending and debt ("spending," "trillion"), immigration ("aliens," "illegal"), and the federal government itself ("government," "bureaucrats").

In particular, separating between Democrats and Republicans allows us to clearly see that the differences go beyond party. Both Republican communities use "Biden" and "border" far more than Democrats do, but these words do not separate the two Republican communities from each other. Rather, between V4 and V3, we note a contrast where the former features more fiscal and anti-government language ("trillion," "debt," "government," "ATF") while the latter features more economic and constituent-oriented language ("financial," "small" business, "community," "proud," "leadership"). Given what is known about the Freedom Caucus and its differences with mainstream Republicans, this tends to align well with how the two tend to differ in priorities and tone more than ideals. For Democrats, we note a similar distinction, where V1 focuses on identity and social-justice

language and V2 focuses more on foreign policy, defense, and district-specific language. Again, we note that they differ in priorities and tones rather than ideals. Overall, the Republican split produces more words with $|z_w| > 1.96$ than the Democratic split (609 versus 377), which, together with the NOMINATE gaps, suggests that the division among Republicans was more divided for the 118th Congress.

### 4.2. Strengths and Limitations

One of the main strengths of our approach is that we decided not to force members into a single group, and instead captured the nuances of representatives' beliefs. The DCMM model and Mixed-SCORE give each member a membership vector, which lets us describe members who sit between factions instead of assigning them to the nearest one.

Combining voting records with speech embeddings also uses two different kinds of behavior, and our speech filter covers nearly the entire House rather than a subset of frequent speakers. For the speech analysis component, we also built in several safeguards where each member counts equally regardless of how much they speak, words are weighted by soft membership, the Fightin' Words $z$-score avoids both common-word and rare-word artifacts, and the within-party comparisons control for party.

Additionally, the communities from $K = 4$ line up with general well-known information. Their ordering on NOMINATE is consistent, and their nearest members correspond to recognizable factions.

Our analysis, however, also has several limitations. First, we note that the our choice of $K = 4$, as mentioned in Section 2.2, was based off of where we saw the eigenvalues begin to drop off. That said, it was not necessarily a formal test and further analysis including additional communities could also provide an even higher-level idea of communities in Congress. Additionally, the weight that $\tau_0 = 1$ was also a modeling choice that we made to give each individual equal relative weights, but the community boundaries may be sensitive to it.

In terms of data, we also know that floor speech is technically a selective record, as members choose when to speak, and 55 members with little floor speech were excluded from the word analysis. Some distinctive words also reflect ceremonial speeches, such as tributes to local teams or officials, rather than policy. A bag-of-words approach also ignores context where, for example, "Biden" can appear in support or criticism.

Additionally, the 118th Congress covered a period with a Republican House majority and a Democratic president. Some vocabulary, such as the focus on the Biden administration, likely reflects this setting rather than stable features of the factions.

### 4.3. Future Work

For further expansion, applying the method to earlier Congresses could allow us to see how things change over time and whether or not these factions are stable over time. It could also be interesting to see whether incumbents move between different groups, specifically on the simplex. Finally, richer text methods using topic models with membership vectors as covariates or analysis of sentiment and framing, could capture differences in how communities discuss the same issue rather than only which words they use. This could add some greater connotative layers to the speech analysis.

## 5. Conclusion

Overall our results suggest that Congress is indeed non-binary. Although party is the dominant structure in the 118th, it is far from the only one. By combining voting records and speech text, we found four communities: progressive Democrats, moderate Democrats, mainstream Republicans, the Freedom Caucus Republicans. Most members tend to lie between these extremes rather than placing most of their weight on a single community. This suggests that a simple binary narrative leaves out nuance in how members of the House vote and speak.

# 6. Appendix

## 6.1. Speech Corpus and Selection

In the 118th Congress, the selection described in the main text reduced the corpus as shown in Table 3. The Record for January 3, 2023 is unavailable from GovInfo. Among the 428 members of the network, 427 have at least one retained speech, 427 have at least five, and 368 (86%) have at least twenty, with a median of 47 speeches per member. For the word analysis, 373 members with at least 1,000 words of floor speech were retained.

| Selection step | Speeches |
|---|---|
| All parsed speeches (House and Senate) | 97,817 |
| House speeches with a member identifier | 46,805 |
| At least 250 characters | 31,568 |
| Dated before November 1, 2024 | 28,914 |
| Excluding Extensions of Remarks | 21,457 |
| By members of the 428-member network (word analysis) | 20,513 |

*Table 3.* Speech selection in the 118th Congress.

## 6.2. Agreement Matrix

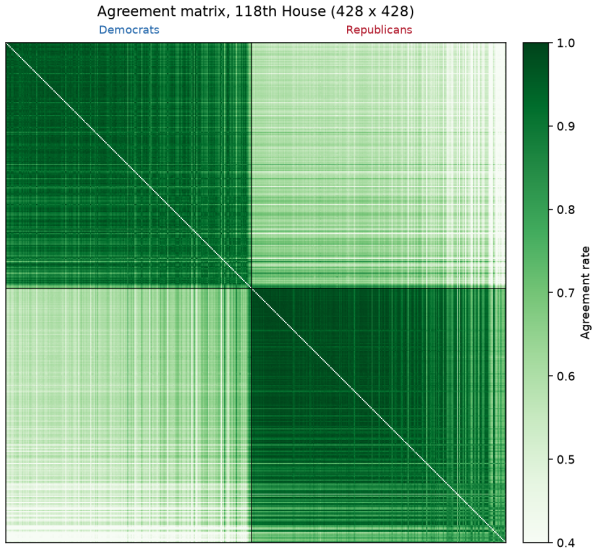



*Figure 1.* Agreement matrix $S$ for the 118th House ($n = 428$).

### 6.3. Scree Plot

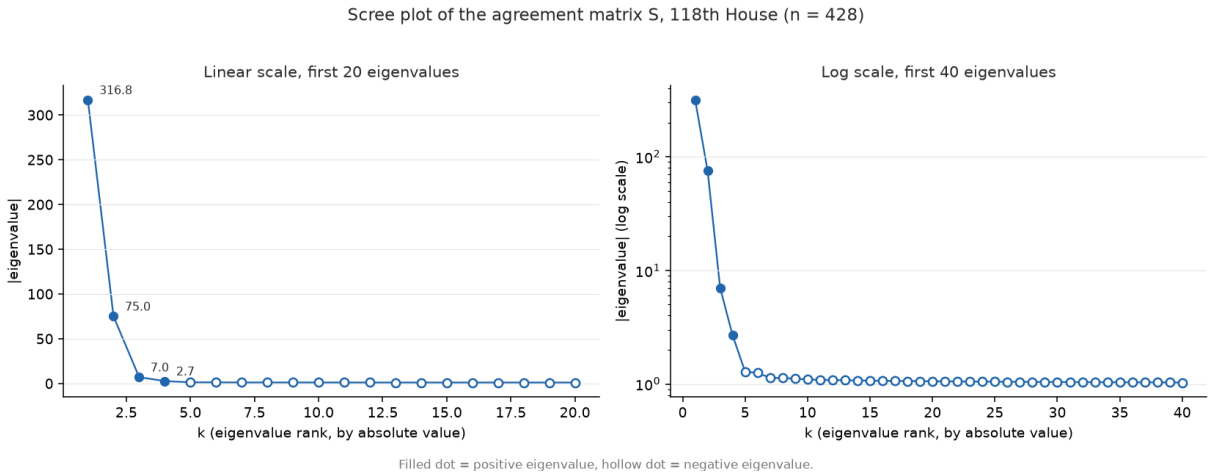



*Figure 2.* Scree plot showing the magnitude of the eigenvalues of $S$.

### 6.4. Left-Right Spectrum

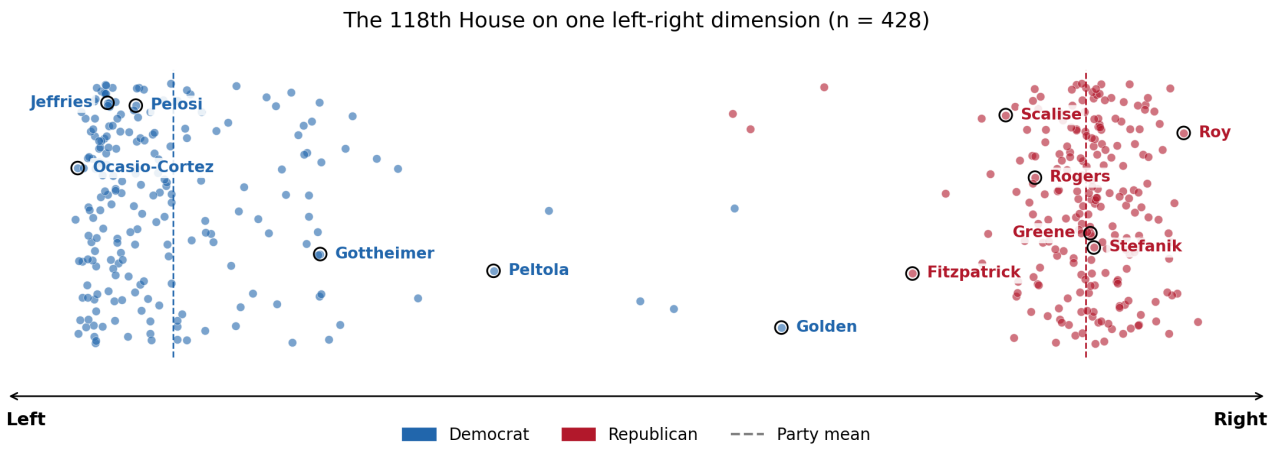



*Figure 3.* Members of the 118th House on one left-right dimension (first principal component of final-passage Yea votes; correlation with DW-NOMINATE dimension 1 of 0.96 on 415 members). Dashed lines mark party means. Labeled members are selected examples.

### 6.5. Membership Tetrahedron

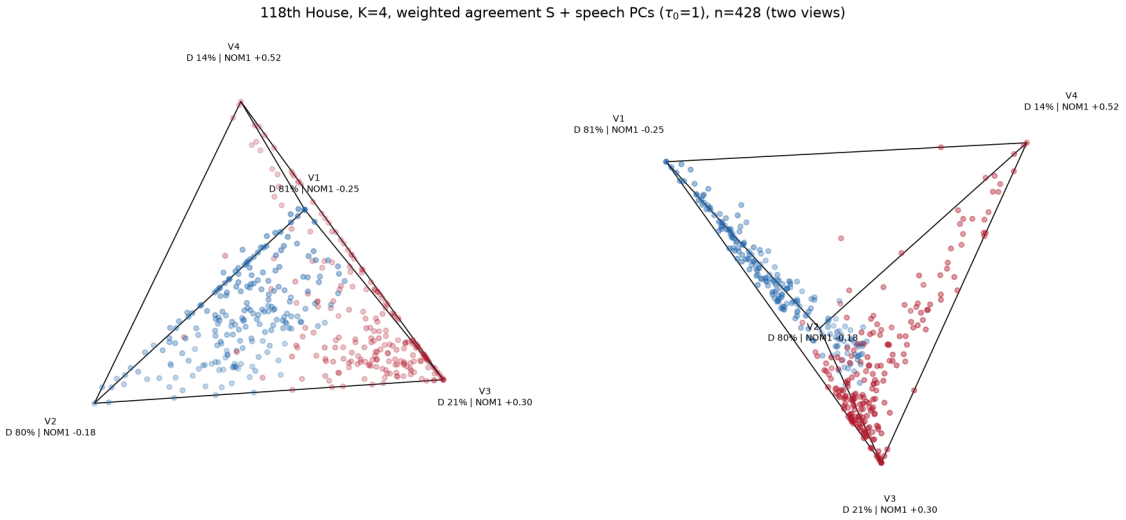



*Figure 4.* Estimated membership vectors $\hat{\pi}_i$ for $K = 4$, shown in a tetrahedron from two viewing angles. Each corner is a community and each point is a member, colored by party.

### 6.6. Word Clouds at the Tetrahedron Corners

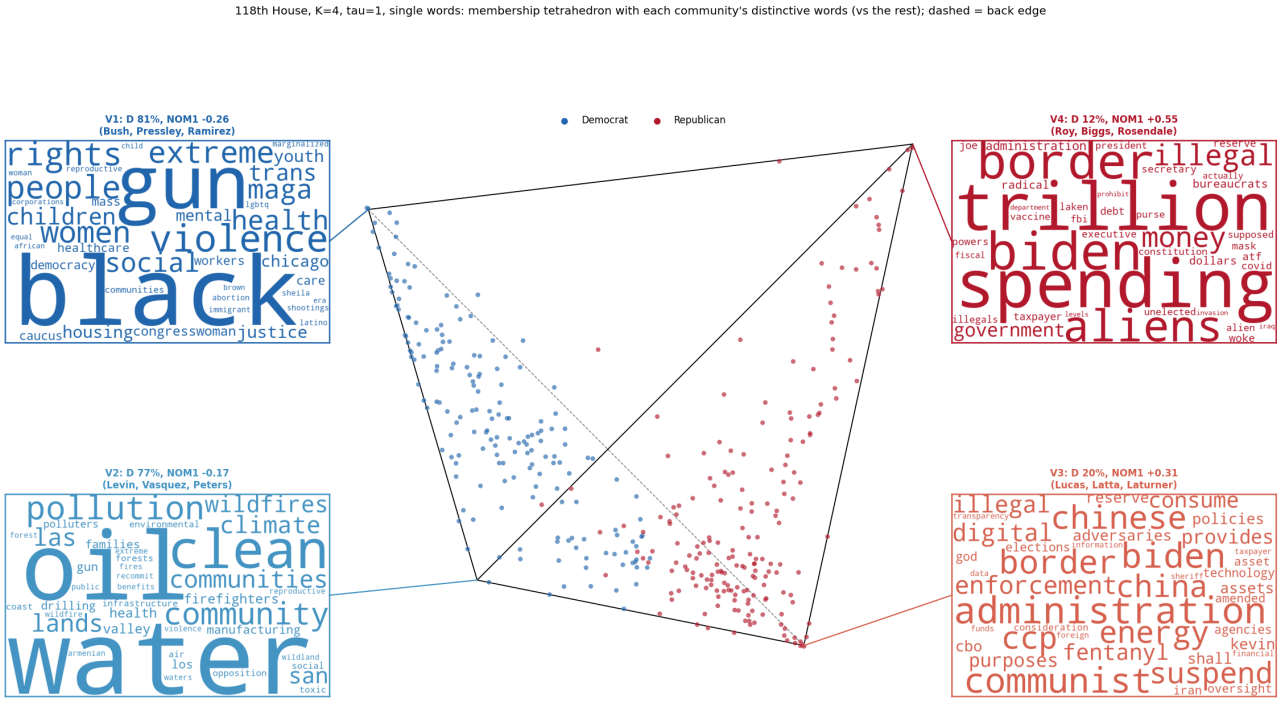



*Figure 5.* The membership tetrahedron for $K = 4$ (two-dimensional projection; the dashed line is the back edge) with each community's 40 most distinctive words relative to the rest of the House. Word size is proportional to $z_w$.

### 6.7. Within-Party Differences

| | Democrats | | Republicans | |
|---|---|---|---|---|
| Rank | V1 over V2 | V2 over V1 | V3 over V4 | V4 over V3 |
| 1 | black (11.7) | las (5.6) | critical (5.7) | trillion (10.7) |
| 2 | people (6.4) | coast (5.5) | financial (5.4) | people (10.5) |
| 3 | rights (5.3) | water (5.4) | proud (5.1) | money (9.0) |
| 4 | justice (5.1) | hamas (5.3) | energy (5.1) | spending (8.9) |
| 5 | child (4.9) | air (5.3) | community (5.0) | government (7.9) |
| 6 | era (4.6) | military (5.3) | small (5.0) | debt (7.6) |
| 7 | trans (4.6) | ukraine (5.1) | communist (4.8) | gender (6.8) |
| 8 | immigrants (4.6) | oil (5.1) | leadership (4.6) | trump (6.6) |
| 9 | chicago (4.4) | trade (4.9) | forward (4.5) | atf (6.5) |
| 10 | housing (4.4) | region (4.6) | families (4.4) | aliens (6.1) |

*Table 4.* Top ten words distinguishing the two main communities within each party ($K = 4$, single words). Values are $|z_w|$.

## 7. Code Availability

Code used in this project can be found here on github: https://github.com/chamlin28/Stat236-Group

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